

Number theory – 7

The integer part, sizes of roots and denominators

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Theoretical minimum. To compute $\lfloor y \rfloor$ one traps y between two consecutive integers; to add integer parts one counts lattice points; and since a non-zero integer is at least 1 in absolute value, one gets lower bounds for $\text{lcm}(1, \dots, n)$ and for denominators.

Definitions. $\lfloor x \rfloor$ is the largest integer not exceeding x , $\{x\} = x - \lfloor x \rfloor$. Put $L(n) = \text{lcm}(1, \dots, n) = \prod_{p \leq n} p^{\lfloor \log_p n \rfloor}$. The order of $\sigma \in S_n$ is the lcm of its cycle lengths (Abstract algebra – 3); Landau's function is $g(n) = \max_{\sigma \in S_n} \text{ord } \sigma$.

Theorem 1 (integer part). (a) If f is continuous, increasing, and $f(x) \in \mathbb{Z}$ only for $x \in \mathbb{Z}$, then $\lfloor f(x) \rfloor = \lfloor f(\lfloor x \rfloor) \rfloor$; e.g. $\lfloor \lfloor x \rfloor / n \rfloor = \lfloor x / n \rfloor$. (b) (Hermite) $\sum_{k=0}^{n-1} \lfloor x + \frac{k}{n} \rfloor = \lfloor nx \rfloor$: the difference of the sides has period $\frac{1}{n}$ and vanishes on $[0, \frac{1}{n})$. (c) (roots) For $a \geq 1, t \geq 0$: $1 + \frac{t}{a} - \frac{(a-1)t^2}{2a^2} \leq (1+t)^{1/a} \leq 1 + \frac{t}{a}$. So $\sqrt[a]{b^a + cb^{a-1}}$ tends to $b + \frac{c}{a}$ from below, and its integer part is found by comparing $b^a + cb^{a-1}$ with an integer a -th power exactly.

Theorem 2 (lattice points). $\sum_{k=1}^n \lfloor f(k) \rfloor$ counts the points $(k, l) \in \mathbb{Z}^2$ with $1 \leq k \leq n, 0 < l \leq f(k)$. For coprime p, q the open diagonal of the rectangle $(0, q) \times (0, p)$ has no lattice points and $(x, y) \mapsto (q-x, p-y)$ swaps its halves, so $\sum_{k=1}^{q-1} \left\lfloor \frac{kp}{q} \right\rfloor = \frac{(p-1)(q-1)}{2}$ (compare Eisenstein's proof in Number theory – 3).

Theorem 3 (Rayleigh–Beatty). If $\alpha, \beta > 1$ are irrational and $\frac{1}{\alpha} + \frac{1}{\beta} = 1$, the sequences $(\lfloor n\alpha \rfloor)_{n \geq 1}$ and $(\lfloor n\beta \rfloor)_{n \geq 1}$ together contain every positive integer exactly once. *Proof.* Exactly $\lfloor N/\alpha \rfloor$ terms $\lfloor n\alpha \rfloor$ are $< N$, and $\lfloor N/\alpha \rfloor + \lfloor N/\beta \rfloor = N - 1$, as the two numbers are non-integers with sum N .

Theorem 4 (Chebyshev-type bounds). $2^n \leq L(n) \leq 4^n$ for $n \geq 7$. *Idea.* If $P \in \mathbb{Z}[x]$, $\deg P < N$, then $L(N) \int_0^1 P \in \mathbb{Z}$; for $P = x^n(1-x)^n$ the integral is in $(0, 4^{-n}]$, so $L(2n+1) \geq 4^n$. By Legendre's formula (Number theory – 4) every prime power in $(k, 2k]$ divides $\binom{2k}{k}$, so $L(2k) \mid L(k) \binom{2k}{k}$; induct. (In fact $\ln L(n) \sim n$, which is the prime number theorem.)

Fact 1. If exactly one of the rationals $r_1, \dots, r_m$ has the least p -adic valuation, their sum has the same valuation — this forces primes into denominators. Landau: $\ln g(n) \sim \sqrt{n \ln n}$ (statement only).

1. Prove Theorem 1(a),(b) and deduce that $\sum_{k \geq 0} \left\lfloor \frac{n+2^k}{2^{k+1}} \right\rfloor = n$ for every positive integer n . Then prove Theorem 2 and show that for arbitrary positive integers p, q with $d = \gcd(p, q)$

$$\sum_{k=0}^{q-1} \left\lfloor \frac{kp}{q} \right\rfloor = \frac{(p-1)(q-1) + d - 1}{2}.$$

2. Prove that $m \binom{n}{m}$ divides $L(n)$ for $1 \leq m \leq n$ by computing the integral $\int_0^1 x^{m-1}(1-x)^{n-m} dx$ in two ways, and deduce Nair's bound $L(n) \geq 2^n$ for $n \geq 7$. Then complete the proof of the upper bound $L(n) \leq 4^n$ of Theorem 4.

3. Let p and q be relatively prime positive integers. Prove that

$$\sum_{k=0}^{pq-1} (-1)^{\lfloor k/p \rfloor + \lfloor k/q \rfloor} = \begin{cases} 0 & \text{if } pq \text{ is even,} \\ 1 & \text{if } pq \text{ is odd.} \end{cases}$$

(IMC 2013)

4. Let a be an even positive integer. Find all real numbers x such that

$$\left\lfloor \sqrt[a]{b^a + x \cdot b^{a-1}} \right\rfloor = b + \left\lfloor \frac{x}{a} \right\rfloor$$

holds for every positive integer b . (IMC 2025)

5. For a positive integer n let $[n] = \{1, 2, \dots, n\}$, let S_n be the set of all bijections from $[n]$ to $[n]$, and T_n the set of all maps from $[n]$ to $[n]$. Define the order $\text{ord}(\tau)$ of $\tau \in T_n$ as the number of distinct maps in the set $\{\tau, \tau \circ \tau, \tau \circ \tau \circ \tau, \dots\}$. Let $f(n) = \max_{\tau \in S_n} \text{ord}(\tau)$ and $g(n) = \max_{\tau \in T_n} \text{ord}(\tau)$ (thus f is Landau's function). Prove that $g(n) < f(n) + n^{0.501}$ for sufficiently large n . (IMC 2025)